

Quantifiers

Quantifiers help us make sense of sentence that is ambiguous by clarifying its scope.

1. Find a way to add a quantifier (or quantifiers) to each sentence below to make it a true statement. Then write the negation of your statement. *If you finish the rest of this activity before we are ready to discuss it as a class, come back and find two more distinct ways to quantify each statement and negate those too.*

For example, $x^2 > 5$ can be quantified

- “For all real numbers $x > \sqrt{5}$, $x^2 > 5$.”
- “For all natural numbers $x \geq 3$...”
- “There exists a real number x such that...”

Those statements can be negated by changing the quantifiers and using our laws for negating different connectives.

- “There exists a real number $x > \sqrt{5}$ such that it is not the case that $x^2 > 5$ ”, or equivalently, “There exists a real number $x > \sqrt{5}$ such that $x^2 \leq 5$.”
- “There exists a natural number $x \geq 3$ ” such that it is not the case that...
- “For all real numbers x , it is not the case that...”

(a) $x^2 > 5 - n$

(b) $x + ay = 3$

(c) $I_n A = A$

2. Many mathematical results can be written as universally quantified implications.
For example, Theorem 1.6 in our textbook can be written logically as

$$\begin{aligned} &\forall A, B \in M_{n \times n} \\ &([(A \text{ is nonsingular}) \wedge (B \text{ is nonsingular})] \\ &\implies [(AB \text{ is nonsingular}) \wedge ((AB)^{-1} = B^{-1}A^{-1})]) \end{aligned}$$

Rewrite the following informally stated results as logically quantified statements in logic notation, then in full sentences.

- (a) Matrix multiplication is associative.
- (b) There's a unique identity matrix for each n .
- (c) The only symmetric and skew symmetric matrix is the zero matrix.
- (d) $A\mathbf{x} = \mathbf{b}$ has a unique solution when A is nonsingular.

3. **Preliminary thinking for WA 02.** The textbook proves the proposition, “For all $n \times n$ matrices A and $m \times 1$ matrices $\mathbf{b}$, if A is nonsingular, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.”

- (a) Rewrite the implication as its contrapositive.
- (b) Rewrite the implication using the law of implication.
- (c) Write the negation of the implication using the law of implication.